Analysis

PENG

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1 Introduction

Definition 1.1 (analysis)**.** *Analysis of real numbers, sequences and series of real numbers, real-valued functions, and geometry [1, 2].*

2 Natural Number

Definition 2.1 (natural number)**.** *A natural number is any element of the set*

$$\mathbb{N} := 0, 1, 2, 3, \dots \tag{2.1}$$

Axiom 2.1 (0)**.** *0 is a natural number.*

Axiom 2.2 (successor)**.** *If n is a natural number, then $n{+}{+}$ is also a natural number.*

Definition 2.2 (notation)**.** $1 := 0{+}{+}$, $2 := 1{+}{+}$, $3 := 2{+}{+}$, ...

Axiom 2.3 ($n{+}{+} \neq 0$)**.** *0 is not the successor of any natural number n.*

Axiom 2.4 ($n{+}{+} \neq m{+}{+}$)**.** *Different natural numbers must have different successors.*
If $n \neq m$, then $n{+}{+} \neq m{+}{+}$. Equivalently, if $n{+}{+} = m{+}{+}$, then $n = m$.

Axiom 2.5 (principle of mathematical induction)**.** *Let $P(n)$ be any property pertaining to a natural number n. Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n{+}{+})$ is also true. Then $P(n)$ is true for every natural number n.*

Lemma 2.1 (recursive definition of sequence)**.** *$a_0 = c$ and $a_{n{+}{+}=f_n(a_n)}$ for each natural number n.*

2.1 Addition

Definition 2.3 (addition)**.** *Define* $0 + m := m$*. Suppose inductively that we have defined* $n + m$*. Then* $(n{+}{+}) + m := (n + m){+}{+}$*.*

$a_n = n + m$ and $f_n(a_n) = a_n{+}{+}$

Lemma 2.2. $n + 0 = n$

Lemma 2.3. $n + (m{+}{+}) = (n + m){+}{+}$

Proposition 2.1 (addition is commutative)**.**

$$n + m = m + n \tag{2.2}$$

Proposition 2.2 (addition is associative)**.**

$$(a + b) + c = a + (b + c) \tag{2.3}$$

Proposition 2.3 (cancellation law)**.** *If* $a + b = a + c$*, then* $b = c$*.*

Definition 2.4. *A natural number* n *is positive if and only if* $n \neq 0$*.*

Lemma 2.4. *If* a *is a positive natural number and* b *is a natural number, then* $a + b$ *is positive.*

Lemma 2.5. *If* $a + b = 0$*, then* $a = 0$ *and* $b = 0$*.*

Lemma 2.6. *If* a *is a positive number, then there exists exactly one natural number* b *such that* $b{+}{+} = a$*.*

Definition 2.5 (order)**.**

- $n \geq m$ *or* $m \leq n$ *if and only if* $n = m + a$ *for some natural number* a
- $n > m$ *or* $m < n$ *if and only if* $n \geq m$ *and* $n \neq m$

Proposition 2.4 (properties of order)**.**

- *reflexive:* $a \geq a$
- *transitive: If* $a \geq b$ *and* $b \geq c$*, then* $a \geq c$
- *antisymmetric: If* $a \geq b$ *and* $b \geq a$*, then* $a = b$
- *addition preserve order:* $a \geq b$ *if and only if* $a + c \geq b + c$
- $a < b$ *if and only if* $a{+}{+} \leq b$
- $a < b$ *if and only if* $b = a + d$ *for some positive number* d

Proposition 2.5 (trichotomy of order)**.** *Exactly one of following statement is true:* $a < b$*,* $a = b$*, or* $a > b$*.*

Proposition 2.6 (strong principle of induction)**.** *Let* m_0 *be a natural number. Suppose that for each* $m \geq m_0$*, if* $P(m')$ *is true for all* $m_0 \leq m' < m$*, then* $P(m)$ *is also true. Then* $P(m)$ *is true for all* $m \geq m_0$*.*

Example 2.1 (Fibonacci sequence)**.**

$$F_0 = 0 \tag{2.4}$$

$$F_1 = 1 \tag{2.5}$$

$$F_n = F_{n-1} + F_{n-2} \tag{2.6}$$

2.2 Multiplication

Definition 2.6 (multiplication). *Define* $0 \times m := 0$. *Suppose inductively that we have defined* $n \times m$. *Then* $(n++) \times m := (n \times m) + m$.

Lemma 2.7. $n \times 0 = 0 \times n$

Lemma 2.8. $n \times (m++) = (n \times m) + n$

Proposition 2.7 (multiplication is commutative).

$$n \times m = m \times n \tag{2.7}$$

Lemma 2.9 (zero divisor). $n \times m = 0$ *if and only if at least one of* n *and* m *is equal to 0. If* n *and* m *are both positive, then* nm *is also positive.*

Proposition 2.8 (distributive law).

$$a(b+c) = (b+c)a = bc + ca \tag{2.8}$$

Proposition 2.9 (multiplication is associative).

$$(ab)c = a(bc) \tag{2.9}$$

Lemma 2.10 (multiplication preserve order). *If* $a < b$ *and* c *is positive, then* $ac < nc$.

Proposition 2.10 (cancellation law). *If* $ac = bc$ *and* c *is positive, then* $a = b$.

Proposition 2.11 (Euclid's division lemma). *Let* n *be a natural number and* q *be a positive natural number. Then there exist natural numbers* m *and* r, *such that* $0 \leq r < q$ *and* $n = mq + r$.

Proof. $0 = 0 \times q + 0$ and $0 \leq r < q$. Suppose that $n = mq + r$ and $0 \leq r < q$. Then $n++ = mq + (r++)$ and $0 \leq r++ \leq q$. If $r++ = q$, $n++ = mq + q = (m+1)q + 0$. If $r++ < q$, $n++ = mq + r++$. □

Definition 2.7 (Exponentiation). *Define* $m^0 := 1$, *in particular* $0^0 := 1$. *Suppose recursively that* m^n *has been defined for some natural number* n, *then define* $m^{n++} := m^n \times m$.

3 Set Theory

Definition 3.1 (set). *Define a set* A *to be any unordered collection of objects. If* x *is an element of* A, *then* $x \in A$, *otherwise* $x \notin A$.

Axiom 3.1 (sets are objects). *If* A *is a set, then* A *is also an object.*

Axiom 3.2 (equality of sets). $A = B$ *if and only if every element of* A *is an element of* B *and vice versa.*

Axiom 3.3 (empty set). *There exists an empty set* $\emptyset$ *which contains no elements.*

Lemma 3.1 (single choice). *If* A *is a non-empty set, then there exists an object* x *such that* $x \in A$.

Axiom 3.4 (singleton and pair sets). *If* a *is an object, then there exists a singleton set* $\{a\}$ *whose only element is* a. *If* a *and* b *are objects, then there exists a pair set* $\{a, b\}$ *whose only elements are* a *and* b.

Axiom 3.5 (pairwise union). *If* A *and* B *are two sets, then there exists a union of* A *and* B *set* $A \cup B$, *which consists of all the elements which belong to* A *or* B *or both.*

Definition 3.2 (subsets). A *is a subset of* B, *denoted* $A \subseteq B$, *if and only if every element of* A *is also an element of* B.

A *is a proper subset of* B, *denoted* $A \subsetneq B$, *if* $A \subseteq B$ *and* $A \neq B$.

Axiom 3.6 (axiom of specification). *Let* A *be a set and* $P(x)$ *be a property pertaining to* x *for each* $x \in A$. *Then there exists a set* $\{x \in A : P(x) \text{ is true}\}$ *whose elements are precisely the elements* x *in* A *for which* $P(x)$ *is true.*

Definition 3.3 (intersections). *The intersection of two sets is defined to be the set*

$$A \cap B := \{x \in A : x \in B\} \tag{3.1}$$

Definition 3.4 (difference sets). *Define $A - B$ or $A \backslash B$ to be set A with elements of B removed*

$$A \backslash B := \{x \in A : x \notin B\} \tag{3.2}$$

Axiom 3.7 (replacement). *If $P(x, y)$ is a property pertaining to x and y, such that for each $x \in A$ there is at most one y for which $P(x, y)$ is true. Then there exists a set $\{y : P(x, y)$ is true for some $x \in A\}$.*

Axiom 3.8 (infinity). *There exists a set $\mathbb{N}$, whose elements are called natural numbers, as well as an object 0 in $\mathbb{N}$, and an object $n{+}{+}$ assigned to every $n \in \mathbb{N}$, such that the Peano axioms hold.*

3.1 Cartesian Product

Definition 3.5 (ordered pair). *If x and y are any objects, define the ordered pair (x, y) to be a new object. Two ordered pairs (x, y) and (x', y') are equal if and only if both their components match*

$$(x, y) = (x', y') \iff x = x' \text{ and } y = y'. \tag{3.3}$$

Definition 3.6 (Cartesian product).

4 Function

Definition 4.1 (function). *Let X, Y be sets, and let $P(x, y)$ be a property pertaining to an object $x \in X$ and an object $y \in Y$, such that for every $x \in X$, there is exactly only one $y \in Y$ for which $P(x, y)$ is true. Then the function $f : X \to Y$ is defined by P on the domain X and codomain (range) to be the unique object $f(x) \in Y$ for which $P(x, f(x))$ is true.*

Definition 4.2 (equality of functions). *Two functions $f : X \to Y$ and $g : X' \to Y'$ are equal if their domains and codomains agree, and $f(x) = g(x)$ for all $x \in X$.*

Definition 4.3 (composition). *Let $f : X \to Y$ and $g : Y \to Z$ be two functions, such that the codomain of f is the same set as the domain of g. Then define composition $g \circ f : X \to Z$ to the function*

$$(g \circ f)(x) := g(f(x)) \tag{4.1}$$

Definition 4.4 (injective). *A function is one-to-one if different elements map to different elements*

$$x \neq x' \implies f(x) \neq f(x') \tag{4.2}$$

Equivalently,

$$f(x) = f(x') \implies x = x' \tag{4.3}$$

Definition 4.5 (surjective). *A function is onto if every element in Y comes from applying f to some element in X.*

$$\text{For every } y \in Y, \text{ there exists } x \in X \text{ such that } f(x) = y. \tag{4.4}$$

Definition 4.6 (bijective). *A function is bijective or invertible if it is both one-to-one and onto.*

4.1 Images

Definition 4.7 (images of sets). *If $f : X \to Y$ is a function from X to Y, and S is a subset of X, define $f(S)$ to be the set*

$$f(S) := \{f(x) : x \in S\} \tag{4.5}$$

Definition 4.8 (inverse images). *If U is a subset of Y, define the set $f^{-1}(U)$ to be the set*

$$f^{-1}(U) := \{x \in X : f(x) \in U\} \tag{4.6}$$

Axiom 4.1 (power set axiom). *Let X and Y be sets. Then there exists a set Y^X which consists of all the functions from X to Y*

$$f \in Y^X \iff f \text{ is a function with domain } X \text{ and codomain } Y. \tag{4.7}$$

Axiom 4.2 (union). *Let A be a set whose elements are sets. Then there exists a set $\bigcup A$*

$$x \in \bigcup A \iff x \in S \text{ for some } S \in A. \tag{4.8}$$

5 Integers

6 Rationals

7 Real Numbers

8 Limits of Sequences

9 Power Series

A Tools

Definition A.1 (implication)**.** *"If P, then Q" = "If not Q, then not P".*

Lemma A.1 (de mongan laws)**.**

- *not "A and B" = "not A or not B"*
- *not "A or B" = "not A and not B"*

References

[1] Terence Tao. *Analysis I.* Vol. 1. Springer, 2022. ISBN: 978-981-19-7261-4.

[2] Terence Tao. *Analysis II.* Vol. 2. Springer, 2022. ISBN: 978-981-19-7284-3.